

Explain compare and contrast wired and wireless sensor networks. Explain different topologies for WSN

1. Compare and Contrast: Wired vs. Wireless

Point of Comparison	Wired Sensor Network	Wireless Sensor Network	
Communication Medium	Uses physical cables	Uses wireless signals (RF)	
Installation	Difficult and time-consuming	Easy and fast deployment	
Cost	High installation cost	Lower installation cost	
Reliability	More reliable and stable	Less reliable due to interference	
Mobility	No mobility	Supports mobility	
Scalability	Difficult to expand	Easily scalable	
Power Supply	Continuous power via cables	Battery-powered	
Maintenance	Difficult due to cables	Easier but needs battery replacement	
Interference	Less affected	More affected by noise and obstacles	
Deployment Area	Suitable for fixed environments	Suitable for remote and dynamic areas	

↓

2. Topologies for WSN in IoT

Topology defines how sensor nodes are arranged and how they communicate with the gateway (base station).

Star Topology

In a star network, every sensor node connects directly to a single central gateway or hub.

- **How it Works:** Nodes act as "slaves" and only talk to the "master" hub.
- **Pros:** Simplest to manage; if one node fails, the rest of the network remains unaffected. It consumes the least power for the sensor because it only sends data one hop.
- **Cons:** The central hub is a **single point of failure**; if it goes down, the entire network dies. Range is limited to the signal reach of the hub.

Mesh Topology

Nodes are interconnected, allowing data to "hop" from one sensor to another until it reaches the gateway.

- **How it Works:** Every node can act as a router for other nodes.
- **Pros: High Redundancy.** If one path is blocked or a node fails, the network automatically reroutes data (self-healing). It can cover massive areas by adding more nodes.
- **Cons:** High complexity and "overhead" (nodes must stay awake to route others' data), which drains batteries faster.

Tree (Cluster-Tree) Topology

A hybrid structure where nodes are grouped into clusters, each with a "Cluster Head" (CH).

- **How it Works:** Simple nodes talk to their CH, and CHs talk to each other or the main gateway in a hierarchical fashion.
- **Pros:** Combines the power efficiency of a star (for leaf nodes) with the range of a mesh. Highly scalable for massive IoT deployments (e.g., smart agriculture).
- **Cons:** If a Cluster Head fails, the entire branch (cluster) is disconnected from the network.

Summary of Selection

- **Choose Wired** for: Industrial control systems, security cameras, or when sensors are in environments with thick concrete/metal that blocks signals.
- **Choose Wireless (Star)** for: Home automation or small offices where all devices are close to a router (WiFi/Zigbee).
- **Choose Wireless (Mesh/Tree)** for: Smart cities, agriculture, and large-scale environmental monitoring where coverage needs to span kilometers.